

Examen du Premier Semestre d'Algèbre**Janvier 2023****Durée : 3h****Pr. BOUETOU B. Thomas****Exercice 1 : (3pts)**

1. Résoudre l'équation suivante :

$$2^x - 3^x = \sqrt{6^x - 9^x} \quad (1.5pts)$$

2. Résoudre l'équation :

$$a^2 + a = 1111111122222222 \quad (1.5pts)$$

Exercice 2 : (3.5pts) Let E and F be two non-empty sets.

1. Let f be a map from E to F . $\forall X, Y \subseteq F$, show that $f^{-1}(X \setminus Y) = f^{-1}(X) \setminus f^{-1}(Y)$. **(1pt)**
2. Show that a map f from E to F is surjective if only if $f(E) = F$. **(1pt)**
3. Show that a map f from E to F is injective if only if $f(A \cap B) = f(A) \cap f(B)$, $\forall A, B \subseteq E$. **(1.5pt)**

Exercice 3 : (4pts) Let $n \in \mathbb{N}^*$. Let us consider the set $U_n = \{z \in \mathbb{C}, z^n = 1\}$.

1. Show that U_n is a subgroup of $\mathbb{C}$ for the multiplication. **(0.75pt)**
2. Show that for all $n, m \in \mathbb{N}^*$, if n divides m then $U_n \subseteq U_m$. **(0.75pt)**
3. Show that if $d = n \wedge m$, then $U_d = U_n \cap U_m$. **(0.75pt)**
4. For $n = 5$, we define $f : \mathbb{Z} \rightarrow \mathbb{C}$ such that $f(k) = e^{i\frac{2k\pi}{5}}$. Show that f is a morphism from the additive group $\mathbb{Z}$ to the multiplicative group U_5 . Determine $\ker(f)$ the kernel of f . **(0.75+0.5pt)**

Problème : (10pts)**Partie A (5pts)***(La suite du problème figure sur la page suivante du sujet original.)*